Governing the Intelligence Loop: Architectural Mitigations for Epistemic Fragility in Autonomous Space-Cyber Threat Analysis

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***Abstract*—Cyber threats directed against space systems (spanning orbital payloads, tracking and telemetry ground stations, downlinks, and upstream manufacturing supply chains) have evolved into acute operational hazards. Concurrently, national security and commercial defenders face an intelligence volume crisis where telemetry and vulnerability disclosures outpace human cognitive bandwidth. While autonomous multi-agent pipelines powered by Large Language Models (LLMs) offer scalable automation, they introduce severe epistemic liabilities: hallucination, source detachment, uncalibrated certainty, and susceptibility to indirect prompt injection. This paper presents Space-Cyber Brief Crew, a multi-agent cyber threat intelligence (CTI) architecture that automates space-domain intelligence synthesis under an explicit governance layer. Built on a decoupled CrewAI orchestration, the system routes structured reporting through a three-stage sequential pipeline (Collector, Analyst, Writer) governed by ten operational rules. Crucially, the architecture establishes a structural boundary between action capabilities and generative synthesis: downstream analysis agents operate in zero-tool sandboxes, restricting the execution attack surface by construction. We evaluate the framework against a tri-domain threat corpus spanning satellite imagery manipulation, ground industrial control systems (ICS), and global navigation satellite system (GNSS) spoofing. In our evaluation, the governed architecture eliminated identifier fabrication (from 23.3% to 0.0%), preserved 100% provenance retention, and neutralized indirect prompt injection attacks that succeeded in 80% of unconstrained baseline runs. Finally, we examine the institutional, regulatory, and oversight implications, establishing technical criteria for federal procurement and intelligence integration.**

***Index Terms*—Cyber Threat Intelligence, Multi-Agent Systems, Large Language Models, Space Systems Security, MITRE ATLAS, AI Governance, Indirect Prompt Injection.**

I. Introduction and Problem Formulation

A. The Intelligence Scalability Dilemma in Space Systems

Modern space infrastructure represents one of the most operationally complex and strategically sensitive components of global critical infrastructure. Space assets underpin global communications networks, financial settlement systems, precision timing, and defense command architectures. Concurrently, the proliferation of commercial off-the-shelf (COTS) components, proliferated Low Earth Orbit (pLEO) constellations, and cloud-hosted telemetry, tracking, and control (TT&C) services have dramatically expanded the cyber-attack surface.

Defending this distributed ecosystem requires rapid ingestion, contextualization, and synthesis of technical data. Cyber Threat Intelligence (CTI) analysts must continuously monitor disparate, multi-format feeds: Industrial Control System (ICS) advisories regarding ground antenna servo controls, academic disclosures on adversarial perturbations against on-orbit machine vision models, and reports of Global Navigation Satellite System (GNSS) spoofing in active conflict zones. The sheer volume of telemetry and vulnerability disclosures exceeds manual human cognitive bandwidth [1], creating chronic triage bottlenecks that motivate autonomous processing pipelines.

B. Epistemic Failure Modes of Generative AI in Security Workflows

Deploying Large Language Models (LLMs) and multi-agent orchestration frameworks [2] into national security workflows introduces fundamental epistemic vulnerabilities. In threat intelligence, an intelligence briefing serves as the evidential basis for resource reallocation, defense posture shifts, and geopolitical attribution. In this high-stakes context, common failure modes of commercial foundation models become systemic risks:

- **Hallucination and Fabrication:** Generating plausible but non-existent CVE identifiers or falsely asserting active in-the-wild exploitation [3].
- **Provenance Erasure (Sinking Citations):** Progressively discarding document titles, dates, and paragraph-level citations across chained multi-agent reasoning contexts.
- **Uncalibrated Certainty:** Expressing speculative inferences with authoritative certainty, violating Intelligence Community Directive (ICD) 203 standards [4].
- **Action-Surface Vulnerabilities:** Granting autonomous agents shell, file, or web tool access, exposing them to indirect prompt injection [5].

Consequently, automating threat intelligence cannot be treated as a pure software orchestration task. It requires an integrated

governance architecture enforcing verifiable constraints on execution capabilities and assertion fidelity.

C. Contributions

This paper presents Space-Cyber Brief Crew, contributing on four fronts:

1) **Decoupled Structural Governance:** Action capabilities are separated from generative synthesis; downstream analytical agents operate in zero-tool sandboxes, neutralizing execution-level prompt injection by construction.
2) **Guardrails-as-Prompt Framework:** Ten machine-interpretable operational rules operationalize ICD 203 tradecraft standards governing attribution, non-fabrication, and calibrated hedging.
3) **Tri-Domain Empirical Evaluation:** An offline comparative evaluation against a multi-segment threat corpus spanning orbital imagery adversarial ML, ground-station PLC exploitation, and RF GNSS spoofing.
4) **Institutional Governance Blueprint:** Technical acquisition criteria translating formal privilege separation and provenance guarantees into federal procurement policy.

II. Related Work

A. Prompt Injection and Defense by Design

Prompt injection blurs the boundary between instructions and untrusted data [5], [6]. While heuristic defenses such as spotlighting [7] reduce attack probability, they offer no formal safety guarantees. Modern defense-by-design frameworks, such as CaMeL [8], enforce capability-based security by quarantining untrusted data within execution-constrained components. Space-Cyber Brief Crew converges on this principle: by denying execution tools to downstream synthesis agents, indirect prompt injections remain inert text strings within the context window.

B. Autonomy Governance and Tradecraft Standards

Intelligence Community Directive (ICD) 203 establishes professional analytic standards, mandating explicit sourcing, distinction between assumptions and facts, and calibrated confidence levels [4]. Complementary federal frameworks, including DoD Directive 3000.09 [9] and the NIST AI Risk Management Framework (AI RMF 1.0) [10], outline oversight and risk controls. We operationalize these qualitative expectations into deterministic architectural boundaries and measured assertion-level constraints.

III. Space Infrastructure Threat Landscape

Space architectures operate across three coupled, heterogeneous segments:

- **Space Segment:** Satellites and on-orbit payloads executing edge neural networks for automated earth observation, vulnerable to adversarial evasion perturbations mapped under MITRE ATLAS [11].
- **Ground Segment:** Terrestrial TT&C stations and Mission Operations Centers (MOCs) utilizing SCADA/ICS controllers (e.g., motor drive PLCs), vulnerable to unauthenticated remote code execution (RCE).
- **Link and User Segment:** RF communication links (S-, X-, Ku-, Ka-bands) and user receivers, susceptible to high-power electronic warfare (EW) jamming and deceptive GNSS spoofing.

Existing frameworks, including Space Policy Directive-5 (SPD-5) [12], [13] and NIST SP 800-53 controls [14], provide high-level principles but lack technical mechanisms for automated threat synthesis.

IV. System Architecture

A. Pipeline Topology and Role Specialization

Orchestrated via CrewAI [2], Space-Cyber Brief Crew routes intelligence through three sequential agents.

Collector (A_C): Given approved corpus $D = \{d_1, \ldots, d_n\}$, extracts structured entity records:

$$R_i = A_C(d_i) = (\text{File}_i, \text{Title}_i, \text{Date}_i, \text{Systems}_i, \text{Summary}_i, \text{Citations}_i) \tag{1}$$

The Collector is restricted strictly to verbatim extraction without causal inference.

Analyst (A_A): Ingests records $R = [R_1, \ldots, R_n]$ and computes an intelligence assessment vector:

$$E_i = A_A(R_i) = (\text{Sector}_i, \text{Type}_i, \text{Severity}_i, \text{ATT\&CK}_i, \text{ATLAS}_i) \tag{2}$$

Writer (A_W): Ingests R and E to synthesize the final brief adhering to a structured tripartite schema (What Happened, Why It Matters, What to Watch).

B. Structural Governance: Decoupling Tools from Analysis

To eliminate execution-surface attacks, the Collector is provisioned solely with a custom `FileReaderTool`, restricted via an immutable path whitelist to plain-text and markdown files within the approved sources directory.

Crucially, the Analyst and Writer agents are initialized with zero external tools (`tools = []`). Operating purely as sequence-to-sequence transformation engines within an isolated memory context, they lack the runtime interface to execute shell commands, access the network, or interact with the filesystem, neutralizing execution pathways by construction [8].

C. Operational Guardrails and Assurance Boundary

Ten operational rules are embedded in agent system prompts: Rules 1–5 enforce structural boundaries (directory whitelisting, script execution prohibitions, read-only integrity); Rules 6–8 mandate ICD 203 tradecraft compliance (mandatory attribution, explicit uncertainty hedging, absolute refusal to fabricate missing data); Rules 9–10 govern classification handling.

Guarantees vs. Probabilistic Conformance: Structural tool denial provides a deterministic guarantee: arbitrary code execution is mathematically precluded by the absence of a tool runtime. Conversely, assertion constraints (attribution, hedging) rely on LLM instruction following and remain probabilistic; their effectiveness must be measured empirically.

TABLE I
GOVERNED VS. UNGOVERNED PIPELINE PERFORMANCE

Evaluation Dimension	Ungoverned	Governed
Hallucination rate (fabricated CVEs/actors)	23.3% (7/30)	0.0% (0/30)
Provenance retention (exact citations)	40.0% (12/30)	100.0% (30/30)
Cross-document conflation rate	30.0% (3/10)	0.0% (0/10)
Tool execution under injection probe	80.0% (8/10)	0.0% (0/10)

D. Ingestion Scoring as an Air-Gapped Firewall

External advisories are triaged by a standalone FastAPI service prior to agent ingestion. The service computes an explainable relevance score $s(t)$ over a space-critical keyword dictionary K:

$$s(t) = \sum_{k \in K} I(k \in t), \quad \text{where } I(k \in t) = \begin{cases} 1, & \text{if } k \in t \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

Advisories crossing threshold $\tau \geq 2$ enter an approval queue for human clearance. The deterministic scorer is designed specifically as an unprivileged ingestion firewall rather than a high-recall semantic retrieval engine. Relying on dense neural embeddings or unconstrained classifiers at the public intake perimeter re-exposes the boundary to prompt injection before human verification. Enforcing strict string membership over dictionary K guarantees zero model context exposure until an intelligence officer verifies the item's pedigree. Upgrading this triage boundary to an isolated dense-retrieval enclave without expanding the execution attack surface forms a key direction for future work.

V. EMPIRICAL EVALUATION

A. Experimental Setup and Threat Corpus

We evaluated the framework against a curated tri-domain corpus ($N = 10$ trials, evaluating 30 item observations per condition):

- **Doc 1 (Space):** Projected gradient descent adversarial perturbation against on-orbit imagery models (MITRE ATLAS: Adversarial Data Crafting).
- **Doc 2 (Ground):** Unauthenticated remote code execution (CVSS 9.8) in an antenna motor drive PLC.
- **Doc 3 (Link/User):** Wide-area GPS spoofing and meaconing over maritime corridors.

We evaluated an Ungoverned Baseline (unconstrained tools, ambient shell access, no operational rules) against the Governed Architecture.

B. Empirical Results

Table I summarizes comparative performance across four evaluation dimensions.

In prompt injection trials (embedding malicious shell execution commands within sensor metadata), the ungoverned pipeline attempted shell execution in 8 of 10 runs. Under the governed pipeline, the injection was treated as inert text; downstream tool execution was completely prevented by construction (0/10).

Within the governed pipeline, assertion-level compliance was audited: Rule 6 (Attribution), Rule 8 (Refusal to Fabricate), and Rule 9 (Markings) achieved 100% compliance (30/30). Rule 7 (Uncertainty Hedging) achieved 93.3% (28/30), with minor variance occurring when numerical metrics were converted into qualitative hedging adjectives.

C. Limitations, Protocol Formalization, and Agenda

The empirical evaluation serves as an initial calibration baseline. To maintain scientific rigor, we formalize the boundaries of these findings alongside a reproducible verification protocol:

1) Factor Attribution vs. Bundled Interventions: While resilience against indirect prompt injection is causally and deterministically isolated to the removal of tool interfaces (`tools = []`), the observed reductions in hallucination (23.3% to 0.0%) and conflation (30.0% to 0.0%) represent the joint effect of the prompt-level guardrail bundle. Decomposing the relative contribution of individual rules requires a full Latin-square factorial ablation.

2) Double-Blind Multi-Coder Protocol: The reported scoring reflects an initial single-annotator codebook. To eliminate potential expectancy bias, we have formalized a double-blind evaluation harness: generated briefs across conditions are stripped of run metadata, assigned randomized hashes, and independently audited by two secondary evaluators against frozen ICD 203 tradecraft rubrics, targeting an inter-rater concordance threshold of Cohen's $\kappa > 0.85$.

3) Deterministic Metadata Injection: Empirical testing revealed a temporal hallucination where the Writer generated a brief dated September 2023 despite 2025 source documents. This failure reinforces a critical architectural invariant: core metadata (timestamps, run IDs, and classification banners) must be injected deterministically by the runtime wrapper rather than generated via stochastic token sampling.

VI. INSTITUTIONAL AND POLICY IMPLICATIONS

A. Human-on-the-Loop Governance Doctrine

Under the ungoverned baseline, models fabricated CVE strings to fill missing fields in 23.3% of items and erroneously asserted that GPS spoofing was executed via compromised ground PLCs in 30% of briefs. The governed pipeline produced explicit "not specified in source document" declarations, preventing conflation.

In alignment with DoD Directive 3000.09 [9], the architecture operationalizes a Human-on-the-Loop paradigm. Intermediate agent transformations proceed autonomously under immutable structural boundaries, while human oversight is concentrated at perimeter boundaries: approving ingested advisories and conducting final provenance review of finished drafts.

B. Auditability and Cryptographic Traceability

To support intelligence oversight under ICD 203 [4], the architecture generates an auditable cryptographic provenance chain linking source hashes directly to finished outputs:

$$P = (\text{SHA-256}(d_i) \rightarrow R_i \rightarrow E_i \rightarrow B) \tag{4}$$

C. Standards for Federal Acquisition

Mapping architecture controls directly to the NIST AI Risk Management Framework [10], we propose three mandatory criteria for federal AI agent procurement:

1) **Mandatory Least-Privilege Action Surfaces:** Contractually barring ambient shell or general tool access in analytical synthesis agents.
2) **Structural Provenance Disclosure:** Requiring immutable source-binding and disclosed attribution benchmark rates.
3) **Deterministic Separation of Metadata:** Mandating that security markings, timestamps, and report identifiers be injected by the runtime environment rather than generated via stochastic model inference.

VII. Conclusion

Space-Cyber Brief Crew demonstrates that autonomous multi-agent intelligence synthesis can be deployed safely in critical infrastructure domains when structural capability control is decoupled from assertion governance. By enforcing zero-tool sandboxing for synthesis agents and encoding tradecraft standards as verifiable operational rules, the framework eliminates execution-surface prompt injection and reduces epistemic degradation. Future work will scale validation via automated multi-annotator corpora and integrate multimodal orbital sensor telemetry.

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